

Edge Geography is XNLP-hard for Pathwidth and in XP for Tree-Partition Width

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Edge Geography

One of the most famous PSPACE-complete problems.

A token starts at a vertex s .

Players alternate:

1. choose an unused edge from the current vertex,
2. move the token along it,
3. delete that edge.

The first player unable to move loses.

Parameterizing the problem

Theorem (Bodlaender, modern language)

(Undirected) Edge Geography is solvable in time $f(k, d) \cdot O(n)$ where k is the treewidth of the graph and d is the maximum degree of any vertex in the graph.

Exponential time becomes "linear time" when two parameters are controlled!

What if we allow d to be unbounded?

Our first result

Theorem

Directed Edge Geography is XNLP-hard parameterized by pathwidth.

XNLP "rules out" fixed-parameter tractability.

Bounded pathwidth $\Rightarrow$ bounded treewidth.

Pathwidth, and thus treewidth, is not enough!

This result transfers to Undirected Edge Geography in a simple way!

Source problem: Chosen Maximum Outdegree

Does there exist an orientation of every edge so that each vertex respects its budget.

$$\sum_{e \in \theta_w(v)} w(e) \leq t(v) \quad \forall v$$

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Known XNLP-complete when parameterized by pathwidth.

Reduction idea in two phases

Phase I — Player 1 encodes an orientation

For each input edge $e = \{x, y\}$, Player 1 chooses one of two sides.

$$\text{choose the } y\text{-side} \iff x \rightarrow y$$

Phase II — Player 2 challenges a vertex

Player 2 chooses x and tests the total weight oriented out of x .

If outgoing weight $> t(x)$, Player 2 has one excursion too many.

Phase 1

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Phase 2

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The full reduction for a small source instance

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Why correctness follows

Feasible orientation

$$\sum_{e \in \theta_\omega(x)} w(e) \leq t(x) \text{ for every } x.$$

Player 2 eventually runs out of non-losing challenges.

Player 1 wins

Infeasible orientation

For some x ,

$$\sum_{e \in \theta_\omega(x)} w(e) > t(x).$$

Player 2 chooses this x and survives one challenge too many.

Player 2 wins

Why pathwidth is preserved

Process an input edge $e = \{x, y\}$ when x and y coexist in the source path decomposition.

Keep only the necessary anchors alive.

Introduce auxiliary gadget vertices locally, then forget them immediately.

$$\text{pw}(G') = O(k)$$

Directed to undirected

Each directed edge is replaced by a constant-size pseudoarc.

$$u \rightarrow v \quad \rightsquigarrow \quad \text{undirected pseudoarc}$$

Safe traversal works from u to v .

Entering the wrong way is losing under optimal play.

Directed hardness transfers to Undirected Edge Geography.

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XP membership for width of treepartition

A rooted tree partition groups vertices into bags.

Edges are either:

- inside one bag, or
- between adjacent bags of the tree.

For width k , a parent-child cut has at most k^2 ports (on simple graphs).

Theorem

Undirected Edge Geography on simple graphs, given together with a rooted tree partition of width k , is solvable in XP time parameterized by k .

Tree-partition

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A rough overview of the algorithm

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A solved child conveys a boolean function

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The child exports a response

$$\Phi_a(\nu_b, \nu_c) = \nu_b \wedge \nu_c$$

ν_b and ν_c are the values of the two possible continuations in the parent.

The cut has $O(k^2)$ ports,

Child counters comprise the worst factor

The n -dependent part

For fixed k , there are t_k possible child types.

Each counter lies in $\{0, \dots, n\}$, so

$$m \in \{0, \dots, n\}^{t_k} \implies \#m \leq (n+1)^{t_k} = n^{O(t_k)}.$$

The other factors

Unused ports and internal edges contribute

$$2^{k^2} \cdot 2^{\binom{k}{2}} = 2^{O(k^2)}.$$

All other boundary data depends only on k ; collect it into $g(k)$.

$$\underbrace{n}_{\text{bags}} \cdot \underbrace{g(k)}_{\text{includes } 2^{O(k^2)}} \cdot \underbrace{n^{O(t_k)}}_{\text{counter states}} = n^{f(k)}.$$

Ridiculously bad in practice

Let T_r be the number of ambient types on an interface with r ports.

ports r	0	1	2	3	4
ambient types T_r	1	2	676	$\approx 2^{768}$	$\geq 2^{2^{4058}}$

Illustrative ambient count obtained by unwinding Definition 35.

The number of realizable types may be vastly smaller.

A width- k cut may have k^2 ports.

The crude bound is a tower of exponentials of height $O(k^2)$.

Takeaway

Pathwidth

small separators

uncontrolled residual history

XNLP-hard

Tree-partition width

bounded interfaces

residual play can be compressed

XP

Is Edge Geography in XP parameterized by pathwidth alone?
Is Edge Geography XALP-hard when parameterized by treepartition width?

Thanks for listening!

Questions welcome!